

KNighter or other
LLM generator

(a) LLM Generator
context

Draft
checker/query



Patch



Source tree



Validation target



Sec. 4
overview



Extensible Refinement Layer for LLM-Based Checker Generation



(b) Evidence assembly (typed)

Patch preflight

- Patch facts
- Rule-guided requirements

Source extraction

semantic_slice: function, guards,
calls, mem/state, build flags

Analyzer evidence

- CSA: CFG / call graph
- Optional checker-execution observations
- CodeQL: source-derived context

Validation feedback

validation_outcome:
failure_mode, repair

Secs. 4.1-4.2

Merge · deduplicate · retain missingness



(c) Evidence Bundle Schema

Typed records and bounded
views

- Origin:
internal / output / source-
derived
- Scope · location · semantic
payload

Interface · raw-output hash
Missing evidence stays explicit

Sec. 4.3

(e) Audit artifacts

- evidence bundle
- analyzer results
- final report
- all attempts and
retained checker

Sec. 4.4
output



Retained D*
Best
accepted
checker



(d) Refinement
(paired-warning refinement)

Refinement agent
Edit latest attempt L

Candidate checker/query

Paired validation

Retain a valid, changed
candidate if:

- vulnerable-side warnings
recover; or
- fixed-side reports decrease
with vulnerable-side warnings
retained.

Sec. 4.4



feedback

Latest code + feedback